

FLEET RULE OF THE ROAD TEST FOR AUG 2017

This test is based on: The International Regulations for Preventing Collisions at Sea (1972) with amendments in force up to December 2009 as detailed in "A Seaman's Guide Pocket Book of The International Regulations for Preventing Collisions at Sea" (BR 453 Supp), ATP1(G) Vol II, NP735 and the Mariners Handbook (NP100).

The test is designed to be given to any Warfare Officer, Deck officer, Officer undertaking bridge watch-keeping duties or Young Officer/Cadet. The test should be administered under written exam conditions, the results should be recorded in the OOW training records for inspection by FOST navigation staff. The exam should take no longer than 1hr.

In this test, any change in the status of the ship or of the environment will be indicated in the script. Rules 1 to 19, 34, 35 and Annex IV must be known instinctively for instant application. A thorough knowledge of the other rules is required. The pass mark is 90%.

PART 1

1. Explain what the Rules say about special rules made by an appropriate authority. **(R1b)(1)**.

2. Nothing in the Rules shall exonerate the CO and/or OOW from the consequences of what? **(R2a)(1)**

3. Define the following terms: "vessel engaged in fishing" and "restricted visibility" as it pertains to the Rules. **(R3 d&I)(2)**

4. Describe a sternlight and give one characteristic that differs from a towing light? **(R21 c&d)(2)**

5. Describe the lights you will expect HMS QUEEN ELIZABETH to display from sunset to sunrise during her transit from Rosyth to Portsmouth? **(R23a)(2)**

6. What provision does the Rules make for the position of navigation lights for HMS QUEEN ELIZABETH as result of her special construction?**(R1e)(1)**

7. How does the Rule describe a proper lookout?**(R5)(1)**

8. Where should a vessel keep her course along a narrow channel?**(R9a)(1)**

9. When and how **may** a vessel required to keep her course and speed take action to avoid collision?**(R17a(ii))(1)**

10. Describe the sound signals made by a vessel not under command **(R35c)(1)**

11. What does the term short blast mean?**(R32b)(1)**

12. If the circumstances of the case admit, how should any action taken to avoid collision be?
(R8a)(1)

13. Describe the lights displayed by a sailing vessel less than 20m underway?**(R25 a or b)(1)**

14. When is a vessel deemed to be overtaking another?**(R13b)(1)**

15. Describe the shapes displayed by a vessel engaged in replenishment underway?**(R27b(ii))(1)**

16. How should any alteration of course and/or speed to avoid collision be should the circumstances of the case permit?**(R8d)(2)**

17. How should a vessel using a Traffic Separation Scheme join the scheme and proceed within the scheme? **(R10b(i)(iii))(2)**

Part 2

As the only Midshipmen borne in HMS ALBION (length 176.8m), your Navigating Officer and OsOW are keen to give you the best training before you attend Collingwood for your NWC. The CO has directed that you must achieve 3 consecutive scores greater than 90% in RoR before recommending you for the course. The Ship is on transit to visit Rotterdam, Netherlands prior to transiting through the Orkneys to participate in Exercise Neptune Warrior, NW of Scotland. You have the privilege to conduct the pilotage into Rotterdam.

18. As part of the planning process you note that the Netherlands is in Region A of the IALA buoyage system. What side of the channel as you approach Rotterdam do you expect starboard lateral marks? **(NP375)(0.5)**

19. The pilot vessel reports on VHF that she is passing the fairway buoy (safe water mark). Describe a safe water mark to include the colour, buoy shape, top mark (if any), light colour and light rhythm when fitted? **(NP375)(2.5)**

20. Visibility is good but vessel traffic begins to increase as you approach confined waters. The CO asks what you determine safe speed to be. Give 6 factors you would consider, if you were not taking radar factors into account? **(R6a)(3)**

21. You hear on VHF that a vessel constrained by her draught is heading outbound to sea. How does the Rules define such a vessel?**(R3h)(1)**

22. You sight the vessel constrained by her draught directly ahead of the ship at 5nm with a steady CPA. What situation is developing and what is required from both vessels in accordance with the Rules?**(R14a)(1)**

23. As you reduce speed to embark the pilot, the vessel constrained by her draught alters her course to starboard to follow her route and give you sea room. What sound signal can she use to indicate this manoeuvre?**(R34a)(0.5)**

24. You recommend to the CO to make a sound signal to indicate your alteration of course to starboard. You also recommend using a flashing light to supplement the whistle signal. What signal will you direct the Yeoman to conduct?(**R34b(i)(0.5)**)

25. You are now committed in the channel and with the good weather comes increased activities of small fast boats and sailing vessels. You brief the CO these vessels are required not to impede your safe passage. What is required of these vessels when required by the circumstance of the case to take action to avoid impeding your safe passage? (**R8f(i)(1)**)

26. The SSD OOW reports one of the sailing vessels displaying a cone apex downwards in her rigging appears to be in difficulty 4c ahead with CPA less than 100 yards. What action is required of you in accordance with the Rules given the limitations of the sailing vessel?(**R2b)(1)**)

27. The remainder of the pilotage is uneventful. Being the 2OOD, you quickly change into your Nos 1 uniform as SSD falls out. One of the duty ET(ME) asks what time you want the anchor lights on. What other times are they required to be displayed other than from sunset to sunrise?
(**R20b)(1)**)

28. The Ship sails after 3 nights alongside and the weather forecast is fog patches enroute. As you come back for the middle watch, visibility is less than a 1000yds and fog lookouts are closed up. Describe the sound signals you will expect to be sounding?(**R35a)(1)**)

29. You receive a VHF call from the Pilot vessel to reduce speed to 6kts for the pilot transfer. What lights do you expect the pilot vessel to display as it transits towards you?(**R29a(i)(ii)(2)**)

30. What sound signals will you expect from a sailing vessel underway?(**R35c**)(1)

31. Will this signal change if the sailing vessel is underway but stopped in the water?(**R35c**)(1)

32. Visibility improves half way through the watch to excess of 5nm. At Red 30 range 3nm, you detect a vessel engaged in fishing other than trawling making way through the water. What lights will you expect to see? (**R26c(i)(iii)**)(2)

33. In determining if risk of collision exists, what factors shall be among those taken into account?(**R7d(i)(ii)**)(1)

34. During the Afternoon watch the next day, you hear the CO and the NO discussing the options of anchoring in the Orkneys and narrate their "spin Spec N ditts" for the area. On anchoring, the CO approves for you to keep an anchor watch. Describe the lights you will display at sunset?(**R30a&c**)(2)

35. You weigh anchor in the forenoon and transit towards the exercise areas. Half an hour before you go off your Afternoon watch, you receive a Navtex message of a vessel sinking 100nm West from your current position. Describe 8 distress signals this vessel could use to indicate she is in distress and requires assistance.(**Annex IV**)(4)

36. The CO decides not to attend to the distress assessing the high possibility of other vessels in that area to respond. You record this in the Ship's log. The NO directs that you prepare a short

brief for the OsOW on the NATO special Rules of the Road in preparation for exercising with NATO units. Which unit will have right of way between you during launching and recovery of landing crafts, a replenishment group and a vessel engaged in the recovery of her aircraft?(**ATP 1(G) Vol II Art 140)(2)**

The Dog watch team take over and you quickly get changed to join in the circuit on the flight deck

Part 2 _____/28

TOTAL _____/50

PART 1

1. Explain what the Rules say about special rules made by an appropriate authority. **(R1b)(1)**.

R1b. Nothing in these Rules shall interfere with the operation of special rules made by an appropriate authority for roadsteads, harbours, rivers, lakes or inland waterways connected with the high seas and navigable by seagoing vessels. Such special rules shall conform as closely as possible to these Rules

2. Nothing in the Rules shall exonerate the CO and/or OOW from the consequences of what? **(R2a)(1)**

R2a. Nothing in these Rules shall exonerate any vessel, or the owner, master or crew thereof, from the consequences of any neglect to comply with these Rules or of the neglect of any precaution which may be required by the ordinary practice of seamen, or by the special circumstances of the case

3. Define the following terms: “vessel engaged in fishing” and “restricted visibility” as it pertains to the Rules. **(R3 d&I)(2)**

R3d. The term “vessel engaged in fishing” means any vessel fishing with nets, lines, trawls or other fishing apparatus which restrict manoeuvrability, but does not include a vessel fishing with trolling lines or other fishing apparatus which do not restrict manoeuvrability.

R3I. The term “restricted visibility” means any condition in which visibility is restricted by fog, mist, falling snow, heavy rainstorms, sandstorms or any other similar causes

4. Describe a sternlight and give one way it differs from a towing light? **(R21 c&d)(2)**

R21.

(c). “Sternlight” means a white light placed as nearly as practicable at the stern showing an unbroken light over an arc of the horizon of 135 degrees and so fixed as to show the light 67.5 degrees from right aft on each side of the vessel.

(d). “Towing light” means a yellow light having the same characteristics as the “sternlight” defined in paragraph (c) of this Rule.

5. Describe the lights you will expect HMS QUEEN ELIZABETH to display from sunset to sunrise during her transit from Rosyth to Portsmouth? **(R23a)(2)**

R23a. A power-driven vessel underway shall exhibit:

(i). a masthead light forward;

(ii). a second masthead light abaft of and higher than the forward one; except that a vessel of less than 50 metres in length shall not be obliged to exhibit such light but may do so;

(iii). sidelights;

(iv). a sternlight

6. What provision does the Rules make for the position of navigation lights for HMS QUEEN ELIZABETH as result of her special construction?**(R1e)(1)**

R1e. Whenever the Government concerned shall have determined that a vessel of special construction or purpose cannot comply fully with the provisions of any of these Rules with respect

to the number, position, range or arc of visibility of lights or shapes, as well as to the disposition and characteristics of sound-signalling appliances, such vessel shall comply with such other provisions in regard to the number, position, range or arc of visibility of lights or shapes, as well as to the disposition and characteristics of sound-signalling appliances, as her Government shall have determined to be the closest possible compliance with these Rules in respect of that vessel

7. How does the Rule describe a proper lookout?(**R5)(1)**

R5. Every vessel shall at all times maintain a proper look-out by sight and hearing as well as by all available means appropriate in the prevailing circumstances and conditions so as to make a full appraisal of the situation and of the risk of collision

8. Where should a vessel keep her course along a narrow channel?(**R9a)(1)**

R9a. A vessel proceeding along the course of a narrow channel or fairway shall keep as near to the outer limit of the channel or fairway which lies on her starboard side as is safe and practicable

9. When and how **may** a vessel required to keep her course and speed take action to avoid collision?(**R17a(ii))(1)**

R17a(ii). The latter vessel may however take action to avoid collision by her manoeuvre alone, as soon as it becomes apparent to her that the vessel required to keep out of the way is not taking appropriate action in compliance with these Rules.

10. Describe the sound signals made by a vessel not under command (**R35c)(1)**

R35c. a vessel not under command, a vessel restricted in her ability to manoeuvre, a vessel constrained by her draught, a sailing vessel, a vessel engaged in fishing and a vessel engaged in towing or pushing another vessel shall, instead of the signals prescribed in paragraphs (a) or (b) or this Rule, sound at intervals of not more than 2 minutes three blasts in succession, namely one prolonged followed by two short blasts.

11. What does the term short blast mean?(**R32b)(1)**

R32b. The term "short blast" means a blast of about one second's duration.

12. If the circumstances of the case admit, how should any action taken to avoid collision be?
(R8a)(1)

R8a. Any action to avoid collision shall be taken in accordance with the Rules of this Part and shall, if the circumstances of the case admit, be positive, made in ample time and with due regard to the observance of good seamanship

13. Describe the lights displayed by a sailing vessel less than 20m underway?**(R25 a or b)(1)**

R25a. A sailing vessel underway shall exhibit:

(i). sidelights;

(ii). a sternlight;

(b). In a sailing vessel of less than 20 metres in length the lights prescribed in paragraph (a) of this Rule may be combined in one lantern carried at or near the top of the mast where it can best be seen.

14. When is a vessel deemed to be overtaking another?**(R13b)(1)**

R13b. A vessel shall be deemed to be overtaking when coming up with another vessel from a direction more than 22.5 deg abaft her beam, that is, in such a position with reference to the vessel she is overtaking, that at night she would be able to see only the sternlight of that vessel but neither of her sidelights.

15. Describe the shapes displayed by a vessel engaged in replenishment underway?**(R27b(ii))(1)**

R27b(ii). three shapes in a vertical line where they can best be seen. The highest and lowest of these shapes shall be balls and the middle one a diamond

16. How should any alteration of course and/or speed to avoid collision be should the circumstances of the case permit?**(R8d)(2)**

R8b. Any alteration of course and/or speed to avoid collision shall, if the circumstances of the case admit, be large enough to be readily apparent to another vessel observing visually or by radar; a succession of small alterations of course and/or speed should be avoided.

17. How should a vessel using a Traffic Separation Scheme join the scheme and proceed within the scheme? **(R10b(i)(iii))(2)**

R10b

(i). proceed in the appropriate traffic lane in the general direction of traffic flow for that lane;

(iii). normally join or leave a traffic lane at the termination of the lane, but when joining or leaving from either side shall do so at as small an angle to the general direction of traffic flow as practicable.

Part 2

As the only Midshipmen borne in HMS ALBION (length 176.8m), your Navigating Officer and OsOW are keen to give you the best training before you attend Collingwood for your NWC. The CO has directed that you must achieve 3 consecutive scores greater than 90% in RoR before recommending you for the course. The Ship is on transit to visit Rotterdam, Netherlands prior to transiting through the Orkneys to participate in Exercise Neptune Warrior, NW of Scotland. You have the privilege to conduct the pilotage into Rotterdam.

18. As part of the planning process you note that the Netherlands is in Region A of the IALA buoyage system. What side of the channel as you approach Rotterdam do you expect starboard lateral marks? **(NP375)(0.5)**

NP375. Starboard side

19. The pilot vessel reports on VHF that she is passing the fairway buoy (safe water mark). Describe a safe water mark to include the colour, buoy shape, top mark (if any), light colour and light rhythm when fitted? **(NP375)(2.5)**

NP375.

Colour: Red and White vertical stripes

Shape: Spherical, pillar or spar with topmark

Top mark: Spherical for pillar and spar shapes

Light colour: White

Light rhythm: Isophase, occulting, one long flash every 10 seconds or Morse code "A"

20. Visibility is good but vessel traffic begins to increase as you approach confined waters. The CO asks what you determine safe speed to be. Give 6 factors you would consider, if you were not taking radar factors into account? **(R6a)(3)**

R6a.

(i). the state of visibility;

(ii). the traffic density including concentrations of fishing vessels or any other vessels;

(iii). the manoeuvrability of the vessel with special reference to stopping distance and turning ability in the prevailing conditions;

(iv). at night the presence of background light such as from shore lights or from back scatter of her own lights;

(v). the state of wind, sea and current, and the proximity of navigational hazards;

(vi). the draught in relation to the available depth of water

21. You hear on VHF that a vessel constrained by her draught is heading outbound to sea. How does the Rules define such a vessel?**(R3h)(1)**

R3h. The term "vessel constrained by her draught" means a power-driven vessel which, because of her draught in relation to the available depth and width of navigable water, is severely restricted in her ability to deviate from the course she is following

22. You sight the vessel constrained by her draught directly ahead of the ship at 5nm with a steady CPA. What situation is developing and what is required from both vessels in accordance with the Rules?**(R14a)(1)**

Head on situation.

R14a. When two power-driven vessels are meeting on reciprocal or nearly reciprocal courses so as to involve risk of collision each shall alter her course to starboard so that each shall pass on the port side of the other.

23. As you reduce speed to embark the pilot, the vessel constrained by her draught alters her course to starboard to follow her route and give you sea room. What sound signal can she use to indicate this manoeuvre?(R34a)(0.5)

R34a. When vessels are in sight of one another, a power-driven vessel underway, when manoeuvring as authorized or required by these Rules, shall indicate that manoeuvre by the following signals on her whistle:

- one short blast to mean "I am altering my course to starboard";
- two short blasts to mean "I am altering my course to port";
- three short blasts to mean "I am operating astern propulsion"

24. You recommend to the CO to make a sound signal to indicate your alteration of course to starboard. You also recommend using a flashing light to supplement the whistle signal. What signal will you direct the Yeoman to conduct?(R34b(i))(0.5)

R34b(i). Any vessel may supplement the whistle signals prescribed in paragraph (a) of this Rule by light signals, repeated as appropriate, whilst the manoeuvre is being carried out:

(i). these light signals shall have the following significance:

- one flash to mean "I am altering my course to starboard";
- two flashes to mean "I am altering my course to port";
- three flashes to mean "I am operating astern propulsion";

25. You are now committed in the channel and with the good weather comes increased activities of small fast boats and sailing vessels. You brief the CO these vessels are required not to impede your safe passage. What is required of these vessels when required by the circumstance of the case to take action to avoid impeding your safe passage? (R8f(i))(1)

R8f(i). A vessel which, by any of these Rules, is required not to impede the passage or safe passage of another vessel shall, when required by the circumstances of the case, take early action to allow sufficient sea-room for the safe passage of the other vessel.

26. The SSD OOW reports one of the sailing vessels displaying a cone apex downwards in her rigging appears to be in difficulty 4c ahead with CPA less than 100 yards. What action is required of you in accordance with the Rules given the limitations of the sailing vessel?(R2b)(1)

R2b. In construing and complying with these Rules due regard shall be had to all dangers of navigation and collision and to any special circumstances, including the limitations of the vessels involved, which may make a departure from these Rules necessary to avoid immediate danger.

27. The remainder of the pilotage is uneventful. Being the 2OOD, you quickly change into your Nos 1 uniform as SSD falls out. One of the duty ET(ME) asks what time you want the anchor lights on. What other times are they required to be displayed other than from sunset to sunrise? (R20b)(1)

R20b. The lights prescribed by these Rules shall, if carried, also be exhibited from sunrise to sunset in restricted visibility and may be exhibited in all other circumstances when it is deemed necessary

28. The Ship sails after 3 nights alongside and the weather forecast is fog patches enroute. As you come back for the middle watch, visibility is less than a 1000yds and fog lookouts are closed up. Describe the sound signals you will expect to be sounding?(R35a)(1)

R35a(i). A power-driven vessel making way through the water shall sound at intervals of not more than 2 minutes one prolonged blast.

29. You receive a VHF call from the Pilot vessel to reduce speed to 6kts for the pilot transfer. What lights do you expect the pilot vessel to display as it transits towards you?(R29a(i)(ii))(2)

R29a. A vessel engaged on pilotage duty shall exhibit:

(i). at or near the masthead, two all-round lights in a vertical line, the upper being white and the lower red;

(ii). when underway, in addition, sidelights and a sternlight;

30. What sound signals will you expect from a sailing vessel underway?(R35c)(1)

R35c. A vessel not under command, a vessel restricted in her ability to manoeuvre, a vessel constrained by her draught, a sailing vessel, a vessel engaged in fishing and a vessel engaged in towing or pushing another vessel shall, instead of the signals prescribed in paragraphs (a) or (b) of this Rule, sound at intervals of not more than 2 minutes three blasts in succession, namely one prolonged followed by two short blasts

31. Will this signal change if the sailing vessel is underway but stopped in the water?(R35c)(1)

No

32. Visibility improves half way through the watch to excess of 5nm. At Red 30 range 3nm, you detect a vessel engaged in fishing other than trawling making way through the water. What lights will you expect to see? (R26c(i)(iii))(2)

R26c. A vessel engaged in fishing, other than trawling shall exhibit:

(i). two all-round lights in a vertical line, the upper being red and the lower white, or a shape consisting of two cones with apexes together in a vertical line one above the other

(iii). when making way through the water, in addition to the lights prescribed in this paragraph, sidelights and a sternlight.

33. In determining if risk of collision exists, what factors shall be among those taken into account?(R7d(i)(ii))(1)

R7d(i)(ii). In determining if risk of collision exists the following considerations shall be among those taken into account:

(i). such risk shall be deemed to exist if the compass bearing of an approaching vessel does not appreciably change;

(ii). such risk may sometimes exist even when an appreciable bearing change is evident, particularly when approaching a very large vessel or a tow or when approaching a vessel at close range.

34. During the Afternoon watch the next day, you hear the CO and the NO discussing the options of anchoring in the Orkneys and narrate their "spin Spec N ditts" for the area. On anchoring, the CO approves for you to keep an anchor watch. Describe the lights you will display at sunset?(R30a&c)(2)

R30a. A vessel at anchor shall exhibit where it can best be seen:

(i). in the fore part, an all-round white light or one ball;

(ii). at or near the stern and at a lower level than the light prescribed in subparagraph (i), an all-round white light.

R30c. A vessel at anchor may, and a vessel of 100 metres and more in length, shall also use the available working or equivalent lights to illuminate her decks

35. You weigh anchor in the forenoon and transit towards the exercise areas. Half an hour before you go off your Afternoon watch, you receive a Navtex message of a vessel sinking 100nm West from your current position. Describe 8 distress signals this vessel could use to indicate she is in distress and requires assistance.(Annex IV)(4)

Annex IV. The following signals, used or exhibited either together or separately, indicate distress and need of assistance:

(a). a gun or other explosive signal fired at intervals of about a minute;

(b). a continuous sounding with any fog-signalling apparatus;

(c). rockets or shells, throwing red stars fired one at a time at short intervals;

(d). a signal made by radiotelegraphy or by any other signalling method consisting of the group ...-- -... (SOS) in the Morse Code;

(e). a signal sent by radiotelephony consisting of the spoken word "Mayday";

(f). the International Code Signal of distress indicated by N.C.;

(g). a signal consisting of a square flag having above or below it a ball or anything resembling a ball;

(h). flames on the vessel (as from a burning tar barrel, oil barrel, etc.);

(i). a rocket parachute flare or a hand-flare showing a red light;

(j). a smoke signal giving off orange-coloured smoke;

(k). slowly and repeatedly raising and lowering arms outstretched to each side;

(l). the radiotelegraph alarm signal;

(m). the radiotelephone alarm signal;

(n). signals transmitted by emergency positioning-indicating radio beacons;

(o). approved signals transmitted by radiocommunication systems, including survival craft radar transponders.

2.. The use or exhibition of any of the foregoing signals except for the purpose of indicating distress and need of assistance and the use of other signals which may be confused with any of the above signals is prohibited.

3.. Attention is drawn to the relevant sections of the International Code of Signals, the Merchant Ship Search and Rescue Manual and the following signals:

(a). a piece of orange-coloured canvas with either a black-square and circle or other appropriate symbol (for identification from the air);

(b). a dye marker.

36. The CO decides not to attend to the distress assessing the high possibility of other vessels in that area to respond. You record this in the Ship's log. The NO directs that you prepare a short brief for the OsOW on the NATO special Rules of the Road in preparation for exercising with NATO units. Which unit will have right of way between you during launching and recovery of landing crafts, a replenishment group and a vessel engaged in the recovery of her aircraft?(**ATP 1(G) Vol II Art 140)(2)**

ATP 1(G) Vol II Art 140.

The following rules apply to Allied naval ships and overrule the IRPCS in the circumstances described:

a. Right of way, priority is in the following order

- *Ships engaged in replenishment (other than VERTREP)*
- *Ships engaged in the launching and recovery of landing craft utilities*
- *Ships engaged in the launching and recovery of aircraft*

The Dog watch team take over and you quickly get changed to join in the circuit on the flight deck

Part 2

_____/28

TOTAL

_____/50